

Partial Derivatives and the Gradient



First and second partial derivatives

- 1 For $f(x, y) = x^3y^2 - 4xy + y$, find:
- a** f_x **b** f_y
- c** f_{xx} **d** f_{xy}
- 2 For $f(x, y) = e^{xy}\sin(x + y)$, find f_x using the product rule.
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The gradient and directional derivatives

- 3 Find the gradient ∇f of $f(x, y) = x^2y + 3y^3$ at the point $(1, 2)$.
- 4 Find the directional derivative of $f(x, y) = x^2 + y^2$ at $(1, 1)$ in the direction of $\mathbf{v} = \langle 3, 4 \rangle$.